

PART A

Circuit A

Total resistance in ohms = 100 ohms
Total current from the power supply = 1.25 A
Current through each resistor = 1.25 A
Voltage drop through each resistor
20 ohms = 25 V
30 ohms = 37.5 V
50 ohms = 62.5 V
Power dissipated through each resistor
20 ohms = 31.25 W
30 ohms = 46.88 W
50 ohms = 78.13 W

Circuit B

1. Total resistance in ohms = 12.5 ohms
2. Total current from the power supply = 10 A
3. Current through each resistor =
20 ohms = 6.25 A
30 ohms = 1.25 A
50 ohms = 2.5 A
4. Voltage drop through each resistor = 1.25 V
5. Power dissipated through each resistor
20 ohms = 781.25 W
30 ohms = 156.25 W
50 ohms = 312.5 W